

I chose an x value of 24 for my $P+x$ version. At this point, the replacements are just absurd enough that the poem no longer quite makes sense, but they become abstract enough that it seems intentional. In the third line of the poem with the last word replaced, the seventh possibility reads “Of the pine-trees crusted with *oil*,” where the 24th rank instead says “*sugar*”, and the 41st “*coal*”. Though it seems more logical that a tree is covered in natural oils, oils don’t fit the poem’s setting, whereas the mention of sugar seems like a metaphor for snow, which does fit the theme of the original poem. But the 41st option “*coal*” doesn’t fit the theme and seems illogical. Given that the act of writing poetry often breaks grammatical writing norms, having a language model fill in what it ranks as the 24th most likely option means that the word will still mostly fulfil grammatical requirements and the flow of the poem. Where lower-ranked replacements seem too literal and higher-ranked ones too illogical, the 24th option seems best. Even if it isn’t the most coherent, it seems to be a stylistic choice suited to the medium.

Additionally, to implement my version of $P+7$ technique on all nouns, I would first manually mark each noun in the input text (in my case I used “_____”). Since GPT-2 is autoregressive and I can only feed it preceding text, I would then put the portion of the text before the first blank into a string and input it as context. Then I would take the seventh most likely replacement and append it to my string, along with the original text preceding the next noun. I would repeat this until no nouns remain, with any trailing text copied over unchanged. Should the string become too long for the context window, the starting portion can be trimmed, since it is less contextually important. This way the context keeps growing, including the model’s own words, and the final result is likely far more cohesive than if the nouns had been replaced independently.